TIPA Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program
(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Smart, Cool, and Connected: IoT Systems for Advanced Zero-Carbon Cooling			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification 1)	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical · Materials		Electrical · Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Energy/ Environment Mechanical Systems	Energy/Environmental Control Equipment	Home Appliances and Electronic Application Devices	Air Conditioning and Heating Machines
Concept of Proposed Technology	The proposed technology will develop an IoT-based system built on next generation AI-ML approaches for the intelligent operation of dynamic passive cooling systems for space conditioning of buildings. Passive cooling systems capture climatic resources (e.g., cool night air) and provide access to ambient heat sinks (e.g. cold night skies) to diminish the need for mechanical cooling, providing exceptionally low-energy and low-carbon conditioning. Operable elements such as window apertures, shading devices, and fans improve the performance of such systems dramatically when controlled with high responsiveness to the building's local climate, weather forecasts, HVAC actions, window heat flux, and indoor temperatures. To optimize such control, this project will develop a device that uses AIML tools			

	(specifically using reinforcement learning) to (i) control operable elements automatically when they are directly (wirelessly) controllable, or (ii) to signal occupants that manual operation is needed.
Rationale for Joint R&D Project	We have developed the foundational AI algorithms for passive cooling and tested these in simulation. While we have successfully demonstrated a related technology in active thermal control using temperature sensors and Raspberry Pi based controllers, this has not been done for passive heating/cooling or hybrid (active + passive) conditioning systems. Realizing the full commercial potential of this technology will require collaboration with an industry partner on a joint R&D project to develop, deploy and field test these algorithms on an IoT system comprising of heat flux sensors and a Raspberry Pi based controller. The goal would be to demonstrate the commercial potential of this device as well as the associated energy savings at scale.
Key Words	1. AI agents for zero-energy buildings 2. IoT based HVAC automation 3. Climate-responsive design

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (5)
R&D Objectives	The following specific technologies will be developed and tested in this project: • A small (4cm^3 or less), wireless device that senses heat flux at an interior window surface and amplifies and filters the signal, binning the results as “high outward,” “low outward,” “low inward”, or “high inward,” and transmits the results to a local base unit. • A base unit (built on a Raspberry Pi) that receives the heat flux information in addition to sensed indoor conditions, occupant input, local weather data and forecasts, and HVAC operation. • An AI-based intelligent agent running on the base unit that integrates the information above to determine the optimal states of window apertures, shading devices, movable insulation, fans, or evaporative coolers. • An algorithm that evaluates the system’s passive cooling performance and uses this information to improve the agent’s future performance.			
R&D Target Specifications	The system should achieve space cooling performance within 80% of the theoretical maximum for three building types (single family residential; multi-family residential; small office) in several climates (including humid subtropical, humid continental, warm-summer Mediterranean, and oceanic), as determined by comparing field performance with building co-simulations in EnergyPlus, Matlab, and Python.			

R&D Plan	**Research approach** This work will be accomplished through a collaboration between researchers at Rensselaer Polytechnic Institute (RPI) and the University of Oregon (UO), in addition to our Korean partner(s). RPI will be the lead institution but will fund the work at UO through a sub-contract. RPI researchers will lead the development of the intelligent control agent and its implementation in Raspberry Pi; UO researchers will evaluate the theoretical maximum performance through building energy modeling; RPI, UO, and Korean researchers will each participate in field testing. In our laboratory studies, these AI approaches have shown robustness to building configuration and portability among climate types. A prototype of this device currently exists and is undergoing laboratory testing and refinement. **Key tasks** 1. **Improve the current intelligent agent**: (i) Add the ability to learn from recent history; (ii) Add the ability to respond to weather forecasts; (iii) Add seasonal awareness; (iv) Test the agent in simulations of the building types and climates that will be investigated in the field. 2. **Refine the prototype window heat flux sensing element:** (i) Enable the element to send a low-power signal to a base station, allowing its cable to be removed; (ii) Develop a thermally robust yet re-positionable contact between the heat flux sensor and window surface; (iii) Reduce the element's size to minimize window obstruction 3. **Implement the AI agent into a Raspberry Pi:** The intelligent agent will be implemented in a Raspberry Pi (Raspi), which will be communicating wirelessly with the heat flux sensor(s). The Raspi would offload more complex AI tasks to the backend (cloud server), and the choice of which tasks to offload would need to be determined. 4. **Refine the human-machine interface:** Evaluate human responses to signals for manual controls in testbeds at RPI, UO, and at any Korean testbeds that may be available, and adjust the prototype interface as indicated. 5. **Evaluate system performance in occupied buildings:** Deploy the new prototype in 1-3 occupied buildings near RPI, near UO, and near our Korean partners if possible. Develop EnergyPlus models of the spaces involved. Monitor indoor air temperatures, mechanical space cooling energy consumption, electricity used to power fans and (if present) evaporative coolers, and occupant actions taken. Compare actual to theoretical performance. **Methodologies** Building on our recent work [Park2023], we will use reinforcement learning (RL) for designing control algorithms that the intelligent agent will implement. The implementation of control algorithms into the Raspi, and its integration with the sensor cloud server, will draw upon our prior experience in designing such a system for active heating/cooling [Gupta2016], including using a Smartphone app for user notification and feedback collection. The team has experience in EnergyPlus simulations using system identification and control algorithms implemented in MATLAB

	and Python [Onyejizu2024, Park2023], as well as evaluation in actual buildings. A similar methodology will be followed for field testing and evaluation but focused on passive systems and scaled up with the help of our Korean partner. **Milestones and Timelines** *Total project duration:* 42 months (including the initial 6-month go/no-go period) **M1.** Create an overall design of the system and identify the devices to be used for the heat flux sensor, the base station, and the cloud server (service). Estimate the total system cost and energy savings. [0-6 months, go/no-go milestone] **M2.** Improving the current RL-based AI control agent to include weather forecasts, seasonal awareness. Evaluation in different building types and climate zones through simulations. [7-18 months] **M3.** Implementing the full system prototype using IoT devices, including heat flux sensors, operable shades and Raspi devices that wirelessly communicate with each other. [19-30 months] **M4.** Field testing of the prototype in multiple locations (climate zones), including eastern (RPI) and western (UO) USA, and Korea (with the help of Korean partner(s)). [31-42 months]

4. Role & Responsibility (R&R)

Expected R&R of Research Institute	The US based research institute comprising of two universities (RPI and UO) will have the following roles and responsibilities: 1) The overall system design towards milestone M1 will be done jointly between both US universities (RPI and UO) and Korean SME partner. 2) Improving the RL based control agent towards milestone M2 will be the sole responsibility of RPI. 3) Implementing the full system prototype using IoT devices, towards milestone M3, will be done jointly between RPI and the Korean SME partner. 4) The field testing of the system will be done jointly between both US universities (RPI and UO) and our Korean SME partner. 5) All publication efforts will be led by the US institutions (RPI and UO), and they will also contribute to the evaluation effort through the utilization of their facilities (Watervliet “building-in-a-room” test facility at RPI, and Passive Systems Testbed in Yasui Hall at UO).
Expected R&R of Korean SME Partner	The Korean SME partner will have the following roles and responsibilities: 1) Collaborate will the US institutions in developing the overall system design toward milestone M1. 2) Implementing the full system prototype using IoT devices, in coordination with RPI. This effort will be replicated by both RPI and the Korean partner. 3) Field testing of the system across locations in Korea to determine the energy savings potential in the climate in Korea and gather feedback from building owners/operators (and other potential customers) to evaluate its commercialization potential.

5. Expected Outcome/Impact

Expected Outcomes and Impact	The expected outcomes of the project include the AI based learning and control algorithms and the IoT system that implements these algorithms for passive cooling control. We expect the cost of each IoT system to be under $100, with under $10 in monthly subscription fees for updates and cloud services, providing a quick return of investment (RoI) for customers. Mechanical cooling is the most rapidly growing building energy end-use worldwide, and inefficiencies in electricity generation and transmission cause nearly 30% of the primary energy to be wasted. Well-controlled shading, window apertures, fans, and (in dry climates) evaporative cooling systems can complement -- or even serve as alternatives to -- mechanical conditioning systems and are extensively utilized in traditional building methods. Supporting the widespread use of such systems has the potential not only to reduce energy demands but to greatly reduce greenhouse gas emissions.

6. References

[Gupta2016] S. K. Gupta, S. Atkinson, I. O'Boyle, J. Drogo, K. Kar, S. Mishra, J. T. Wen, BEES: Real-Time Occupant Feedback and Environmental Learning Framework For Collaborative Thermal Management in Multi-Zone, Multi-Occupant Buildings, *Energy and Buildings*, Volume 125, August 2016, Pages 142-152.

[Onyejizu2024] J. Onyejizu, S. Mishra, and K. Kar. Modeling the impact of passive ventilation systems on multi-zone thermal dynamics. In *Proceedings of the 11th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation*, BuildSys '24, page 174–178, New York, NY, USA, 2024.

[Park2023] Park, B., Rempel, A.R. and Mishra, S. Performance, robustness, and portability of imitation-assisted reinforcement learning policies for shading and natural ventilation control. *Applied Energy*, *347*, p.121364, 2023.

☐ Terms:

1) Industrial Technology Classification

Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization